CausalPathNet: A Federated Causal Graph-Augmented Reinforcement Learning Framework for Proactive Patient Journey Orchestration and Counterfactual Treatment Planning

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Abstract

We present CausalPathNet, a novel federated causal graph-augmented reinforcement learning (FCGRL) framework that reframes proactive patient care as a multi-granularity sequential decision problem over causal temporal health graphs. Three principal innovations distinguish our approach from prior healthcare AI platforms: (i) a Dynamic Causal Health Graph (DCHG) encoder that represents patient state as a temporally evolving directed acyclic graph of clinical variables processed via causal graph attention networks (CGAT), enabling structural reasoning over biomarker interdependencies; (ii) a hierarchical RL treatment planner with separate macro-controllers for disease-trajectory management and micro-controllers for daily-intervention scheduling, resolving the long-standing credit assignment problem in clinical decision support; and (iii) a privacy-preserving federated training architecture that coordinates learning across heterogeneous hospital networks without raw data transfer, coupled with a counterfactual treatment simulator for offline evaluation of unseen care pathways. Evaluated on three large-scale longitudinal EHR datasets totaling 2.4 million patient episodes, CausalPathNet achieves 94.1% early disease detection accuracy, reduces avoidable readmissions by 31.4%, and outperforms state-of-the-art clinical AI baselines by up to 16.8 percentage points on composite care outcome metrics.

Keywords-patient journey modeling, causal graph networks, hierarchical reinforcement learning, federated learning, counterfactual treatment planning, electronic health records, explainable AI, clinical decision support

I. Introduction

Contemporary healthcare delivery operates largely in reactive mode: clinical systems diagnose illness after onset, prescribe treatment at discrete encounters, and monitor patients in disconnected silos. This fragmented paradigm generates preventable harm — delayed diagnoses, avoidable readmissions, and suboptimal treatment sequencing account for an estimated 15-30% of total healthcare expenditure in high-income countries [1]. Artificial intelligence offers a structural solution, yet existing clinical AI tools share a common limitation: they treat patient state as a static snapshot rather than a dynamic causal trajectory.

Early work on clinical risk prediction using logistic regression [2] and gradient-boosted trees [3] demonstrated that machine learning could outperform clinical heuristics on isolated prediction tasks. Transformer-based models such as BEHRT [4] and Med-BERT [5] extended this to sequential EHR modeling, capturing temporal patterns across visits. Most recently, reinforcement learning (RL) has been applied to treatment planning in sepsis management [6] and insulin dosing [7], demonstrating the value of sequential decision frameworks. However, none of these approaches simultaneously addresses three critical gaps: (a) causal structure in clinical variable relationships, (b) hierarchical granularity of treatment decisions, and (c) privacy-preserving multi-institutional learning.

CausalPathNet addresses all three with the following contributions:

- **C1. Dynamic Causal Health Graphs (DCHG)** — patient state encoded as a temporal directed acyclic graph with causal edges learned from EHR co-variation patterns and domain ontologies.
- **C2. Hierarchical RL Treatment Planner** — macro-controller managing disease trajectory milestones and micro-controller scheduling daily interventions with separate reward streams.
- **C3. Federated Causal Training (FCT)** — differential privacy-aware federated aggregation across hospital nodes with no raw data exchange.
- **C4. Counterfactual Treatment Simulator (CTS)** — offline evaluation of alternative care pathways via structural causal models and do-calculus intervention operators.

Together, these innovations shift clinical AI from reactive prediction to proactive, causally grounded, privacy-respecting care orchestration.

II. Related Work

A. Clinical Risk Prediction

Classical risk scores (APACHE II, SOFA) use hand-crafted features and logistic regression, performing well in narrow ICU contexts [2]. Deep learning models — LSTMs [8], temporal convolutional networks [9], and transformer architectures [4], [5] — have demonstrated superior sequence modeling but operate as black boxes without causal interpretability, limiting clinical adoption.

B. RL for Clinical Decision Support

Komorowski et al. [6] demonstrated that RL-derived sepsis policies could outperform clinician decisions on retrospective data. Raghu et al. [7] applied actor-critic methods to hypotension treatment. These approaches use flat state representations and single-level policies, lacking the hierarchical structure needed to simultaneously plan across daily interventions and long-term disease trajectories.

C. Federated Learning in Healthcare

Federated learning (FL) has been applied to medical image analysis [10] and mortality prediction [11], enabling multi-site training without data sharing. HealthNet [12] and FedEHR [13] extend FL to EHR data. However, no prior work integrates FL with causal graph state encoders or hierarchical RL policies in a unified framework.

D. Counterfactual Reasoning in Medicine

Pearl's do-calculus [14] provides a theoretical foundation for counterfactual treatment evaluation. CEVAE [15] and GANITE [16] estimate individual treatment effects from observational data. CausalPathNet extends these approaches to multi-step sequential treatment counterfactuals within an RL policy evaluation framework, a combination not previously explored.

III. Problem Formulation

We formalize the proactive patient care problem as a federated hierarchical MDP $M = (S, A, T, R, G, \gamma, F)$ where:

S Graph-structured patient state space. Each state s_t is a DCHG $G_t = (V_t, E_t)$ where nodes are clinical variables (vitals, labs, medications, diagnoses) and directed edges encode causal dependencies learned from observational data.

A Factored action space $A_{macro} \times A_{micro}$. Macro-actions = {Monitor, Intervene, Escalate, Discharge}; micro-actions = {Adjust Medication, Order Lab, Schedule Follow-up, Lifestyle Guidance, Alert Clinician}.

T Stochastic graph-to-graph transition $T(s_{t+1} \mid s_t, a_t)$ learned as a conditional causal generative model.

R Hierarchical reward: clinical improvement (macro) and adherence/safety signals (micro), detailed in Section IV-C.

G Sub-goal space for the macro-controller defining clinical milestones: disease stabilization, biomarker normalization, and safe discharge.

F Federation set of K hospital nodes $\{H_1, ..., H_K\}$ each holding local EHR data under privacy constraints.

The federated objective maximises expected cumulative reward across all patients and hospital nodes:

$$J(\pi) = (1/K) \Sigma_{k=1}^{K} E_{\pi,H_k} [\Sigma_{t=0}^{T} \gamma^t R(s_t, a_t)]$$

IV. CausalPathNet Architecture

A. Dynamic Causal Health Graph (DCHG) Encoder

The DCHG encoder transforms a patient's longitudinal EHR record into a temporally evolving directed acyclic graph at each clinical timestep. Graph construction proceeds in two stages. First, a clinical ontology graph (SNOMED-CT, ICD-10 co-occurrence) provides a structural prior over variable relationships. Second, a differentiable causal discovery module [17] refines edge weights using patient-specific observational co-variation, enabling personalized causal structures.

A three-layer Causal Graph Attention Network (CGAT) — an extension of GAT [18] with directionality-aware attention masks — processes G_t to produce a graph embedding $h_t \in \mathbb{R}^d$:

$$h_t = CGAT(G_t; \Theta_{enc}, M_{causal})$$

where M_{causal} is the causal adjacency mask ensuring attention flows only along valid causal directions, preventing spurious correlation-driven feature mixing that plagues standard GNNs in clinical settings.

B. Hierarchical RL Treatment Planner

The hierarchical planner decomposes treatment decisions across two temporal granularities following the Options framework [19]. The macro-controller π_{macro}, parameterized by a Double Dueling DQN, selects a disease management option o_k upon detecting a trajectory inflection point (defined as a significant shift in DCHG topology). This option specifies a clinical sub-goal g_k — e.g., normalize HbA1c below 6.5% within 90 days.

The micro-controller π_{micro}, conditioned on g_k, executes daily intervention actions via a separate Soft Actor-Critic (SAC) agent, enabling stochastic policies better suited to the continuous-action nature of dosing and scheduling decisions. Crucially, the two controllers use independent replay buffers: the macro-controller receives sparse clinical milestone rewards (weeks-to-months horizon), while the micro-controller receives dense daily adherence and safety rewards, resolving the reward dilution problem that hampers single-level clinical RL.

C. Hierarchical Reward Architecture

Table I: Hierarchical Clinical Reward Structure

Clinical Event	Reward	Controller
Disease remission / stabilization	+2.0	Macro
Biomarker normalization milestone	+1.5	Macro
Safe discharge	+1.0	Macro
Medication adherence (daily)	+0.4	Micro
Lifestyle guideline followed	+0.3	Micro
Adverse drug event detected	-1.5	Macro + Micro
Avoidable readmission	-2.0	Macro
Missed critical alert	-1.0	Micro

D. Federated Causal Training (FCT)

CausalPathNet trains across K hospital nodes without sharing raw patient records. Each hospital node H_k runs local DCHG encoder and policy training on its private EHR shard, computing parameter gradients $\Delta\Theta_k$. The central coordinator aggregates gradients via a causal-structure-aware FedAvg variant:

$$\Theta_{t+1} = \Theta_t - \eta \Sigma_{k=1}^{K} (n_k/n) \Delta\Theta_k + \varepsilon_{DP}$$

where n_k is the patient count at hospital k, n is the total federated population, and ε_{DP} is calibrated Gaussian noise providing (ε, δ)-differential privacy guarantees. Causal structure masks M_{causal} are exchanged in anonymized aggregate form, preserving local ontological nuances while enabling global causal knowledge transfer.

E. Counterfactual Treatment Simulator (CTS)

A central limitation in clinical RL is evaluating policies without live patient exposure. The CTS addresses this via a structural causal model (SCM) M_{SCM} learned over the DCHG transition dynamics. Given an observed patient trajectory $\tau = (s_0, a_0, ..., s_T)$, the CTS generates counterfactual trajectories $\tau\blacksquare$ corresponding to alternative treatment sequences by applying the do-calculus intervention $P(Y \mid do(A = a\blacksquare))$ through abduction-action-prediction steps [14].

This enables clinical stakeholders to query: *What would have happened if this patient had received early insulin therapy instead of lifestyle intervention?* — a question unanswerable by observational models alone. The CTS is validated against randomized trial sub-cohorts where ground-truth counterfactual outcomes are available.

V. Causal Explainability Module

Clinical AI requires interpretability beyond feature importance. CausalPathNet provides three levels of explanation:

L1. Variable-level causal attribution — for each risk prediction, SHAP values are computed over CGAT node embeddings, annotated with causal direction (upstream vs. downstream biomarker).

L2. Trajectory-level narrative — the macro-controller decision path is rendered as a clinical milestone timeline with probability confidence intervals.

L3. Counterfactual contrastive explanation — for each high-risk alert, the system presents the nearest counterfactual trajectory that avoids the adverse outcome, providing actionable clinical rationale.

This three-tier explainability stack directly addresses the "black box" barrier that has historically impeded AI adoption in safety-critical clinical workflows.

VI. Experimental Evaluation

A. Datasets

We evaluate on three longitudinal EHR datasets: (1) *MIMIC-IV* — 383,220 ICU admissions from Beth Israel Deaconess Medical Center; (2) *eICU Collaborative Research Database* — 200,859 episodes across 208 US hospitals; and (3) *UK Biobank Clinical Subset* — 1.82 million primary care episodes. All datasets are de-identified per HIPAA Safe Harbor provisions. The federated setup partitions MIMIC-IV and eICU into hospital-level shards to simulate multi-institutional deployment.

B. Baselines

We compare against: (i) APACHE II clinical scoring, (ii) Random Forest (RF), (iii) Gradient Boosting (XGBoost), (iv) BEHRT [4], (v) Med-BERT [5], (vi) RL-Sepsis [6], (vii) FedEHR [13], and (viii) a standard (non-federated, flat-state) DQN baseline. All deep baselines are tuned with identical compute budgets and evaluated on the same held-out test partitions.

C. Performance Results

Table II: Performance Comparison Across All Baselines

Model	Detection Acc. (%)	AUC-ROC	Readmit. Red. (%)
APACHE II	67.4	0.701	--
Random Forest	76.2	0.789	9.1
XGBoost	79.8	0.821	11.4
BEHRT	84.3	0.863	16.2
Med-BERT	85.9	0.877	18.5
RL-Sepsis	87.1	0.889	21.3
FedEHR	88.6	0.901	23.8
Flat DQN	90.4	0.916	25.7
CausalPathNet	**94.1**	**0.958**	**31.4**

CausalPathNet achieves state-of-the-art performance across all three metrics. The 3.7 pp accuracy gain over Flat DQN directly quantifies the contribution of causal graph encoding. The 31.4% reduction in avoidable readmissions — versus 25.7% for Flat DQN — demonstrates that the hierarchical reward structure better optimizes for long-horizon clinical outcomes rather than visit-level proxies.

D. Ablation Study

Table III: Component Ablation Study

Variant	Acc. (%)	AUC	Readmit. Red. (%)
w/o DCHG (flat state)	90.4	0.916	25.7
w/o causal mask in CGAT	91.7	0.929	27.1
w/o hierarchical RL	92.3	0.936	28.4
w/o federated training	92.9	0.941	29.2
w/o CTS module	93.4	0.949	30.1
Full CausalPathNet	**94.1**	**0.958**	**31.4**

The ablation confirms that every proposed component contributes. The DCHG encoder provides the largest gain (+3.7 pp accuracy), validating the central thesis that causal graph structure encodes clinically meaningful information absent from flat feature vectors. The causal mask in CGAT contributes an additional +1.3 pp by preventing spurious cross-biomarker attention. Federated training adds +1.2 pp by expanding the effective training population while maintaining privacy.

E. Federated Privacy Analysis

We evaluate the privacy-utility tradeoff across differential privacy budgets $\varepsilon \in \{0.5, 1.0, 2.0, 5.0, \infty\}$. At $\varepsilon = 1.0$ (strong privacy), CausalPathNet achieves 93.2% detection accuracy — only 0.9 pp below the non-private federated variant — while providing formal $(\varepsilon, 10^{-5})$-DP guarantees. This compares favorably to FedEHR [13], which requires $\varepsilon > 3.0$ to maintain competitive accuracy, demonstrating the efficiency advantage of our causal-structure-aware aggregation.

VII. Discussion

A. Clinical Significance

The 31.4% reduction in avoidable readmissions translates directly to patient outcomes and cost reduction. Assuming a conservative readmission cost of $15,000 per episode [1] and an annual rate of 3 million preventable readmissions in the US, CausalPathNet-scale deployment could prevent approximately 940,000 readmissions, representing $14.1 billion in annual savings. More importantly, the counterfactual explainability module enables clinicians to understand and trust model recommendations, a prerequisite for safe real-world adoption.

B. Why Causal Graphs Outperform Correlation Models

Standard GNNs and transformers learn correlational patterns that are brittle under distribution shift — e.g., a model trained on pre-pandemic EHR data may misattribute COVID-related biomarker patterns to underlying chronic disease. The DCHG encoder's causal mask enforces invariant causal structure, making learned representations more robust to confounding shifts in hospital protocols and patient demographics across federated nodes.

C. Limitations and Future Directions

Three limitations merit acknowledgment. First, causal structure discovery from observational EHR data is computationally intensive; future work will explore amortized causal inference for real-time deployment. Second, the current framework assumes discrete timesteps aligned with clinical visits; continuous-time extensions using neural ODEs are planned. Third, validation on prospective randomized trials is essential before clinical deployment, as retrospective EHR evaluation cannot fully account for causal confounders in treatment assignment.

VIII. Conclusion

We presented CausalPathNet, a federated causal graph-augmented hierarchical reinforcement learning framework that reconceptualizes proactive patient care as a causally grounded multi-granularity sequential decision problem. By encoding patient state as dynamic causal health graphs, separating disease-trajectory and daily-intervention policy learning, training across hospital networks with differential privacy guarantees, and enabling counterfactual treatment simulation, CausalPathNet achieves 94.1% early disease detection accuracy and reduces avoidable readmissions by 31.4% — state-of-the-art results that reflect a fundamental shift from reactive clinical prediction to proactive, causally interpretable, privacy-preserving care orchestration.

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